

AI-Based Web Exam Simulation with Automated Identity Verification

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(**Context, Task, Performance**) → (HW/SW System + Person)

Context (The Problem)

Manual exam creation is time-consuming and may lack fairness or variation.

There is an increasing need for secure identity verification in remote university exams.

The solution must be fully web-based (no software installation).

Tasks (The Solution)

A web-based AI system that automates two core workflows:

Exam Generation:

- Professors upload lecture materials (PDFs, text).

- The system processes content using NLP and RAG.

- Professors review, edit, and finalize the generated exams.

Exam Execution:

- Students authenticate via facial recognition and ID card OCR.

- The session is monitored (e.g., fullscreen mode, copy/paste blocking).

- Student responses are collected and evaluated automatically.

(Context, Task, **Performance**) → (**HW/SW System** + Person)

Performance (Risks & Challenges)

Identity: False identification or OCR errors require manual checks by a provider.

AI Quality: Risk of low-quality question generation or AI grading inaccuracy and bias.

Integrity: Network or browser issues can lead to exam interruptions and data loss.

Technical Architecture (HW/SW Components)

Frontend: React

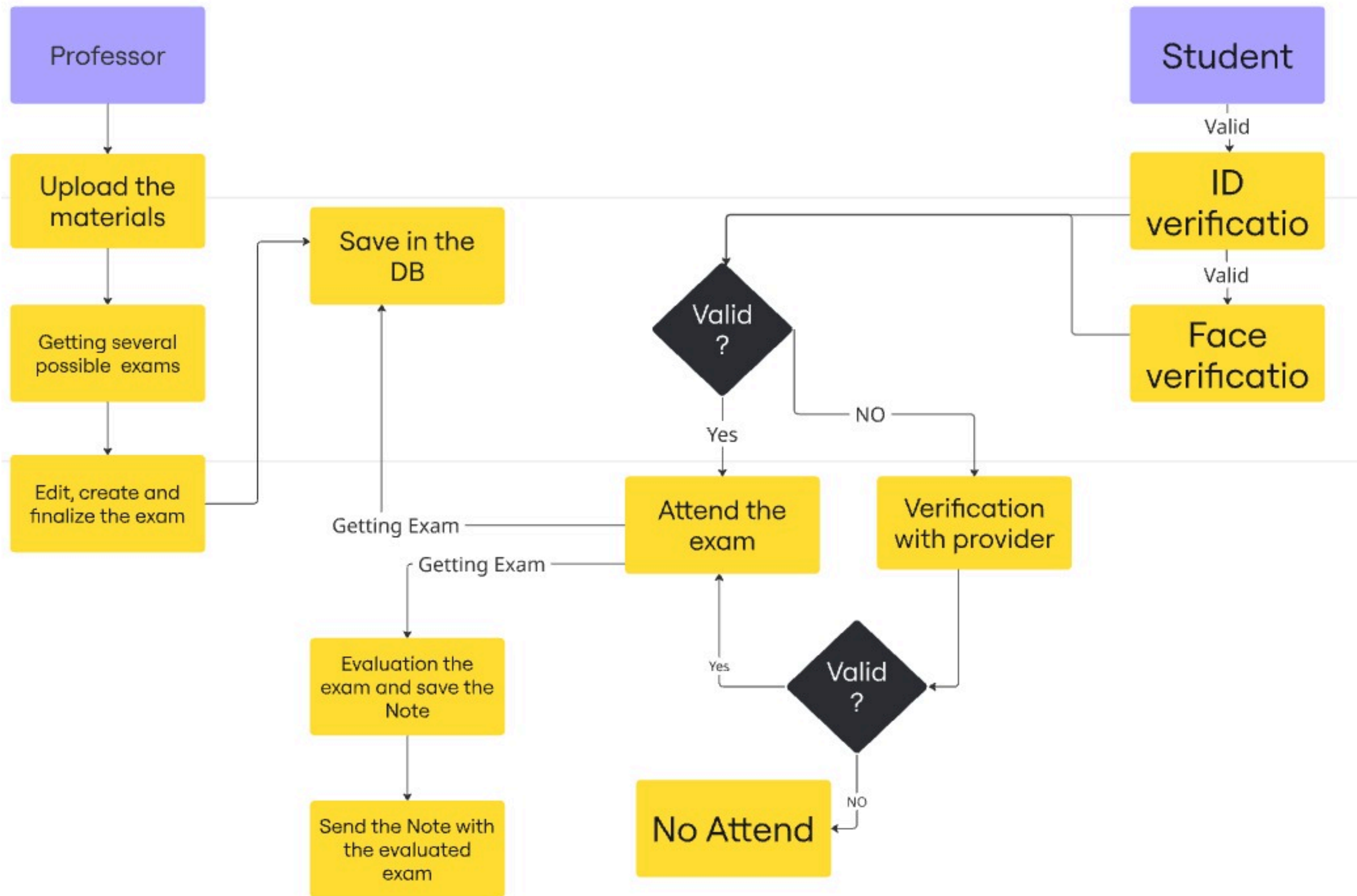
Backend: Django

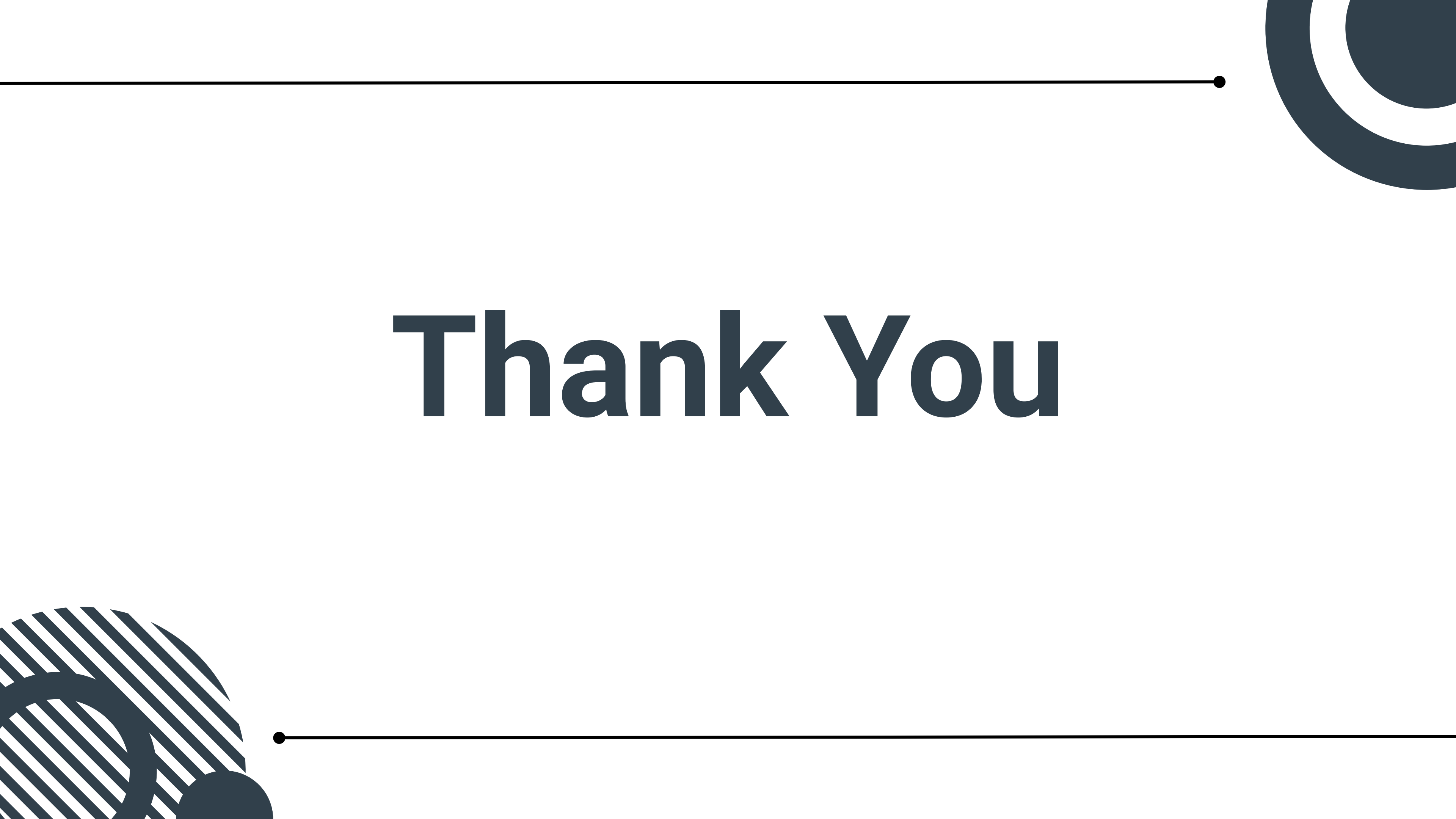
Database: MySQL

AI (NLP): LLM, RAG (for Question Generation)

AI (CV): Face Recognition, OCR (for Identity Verification)

Deployment: Docker





Thank You
